# BASIC METHODS OF FORCE MEASUREMENT

# An unknown force may be measured by the following means:

1. Balancing it against the known gravitational force on a standard mass, either directly or through a system of levers

2. Measuring the acceleration of a body of known mass to which the unknown force is applied

3. Balancing it against a magnetic force developed by interaction of a current carrying coil and a magnet

4. Transducing the force to a fluid pressure and then measuring the pressure

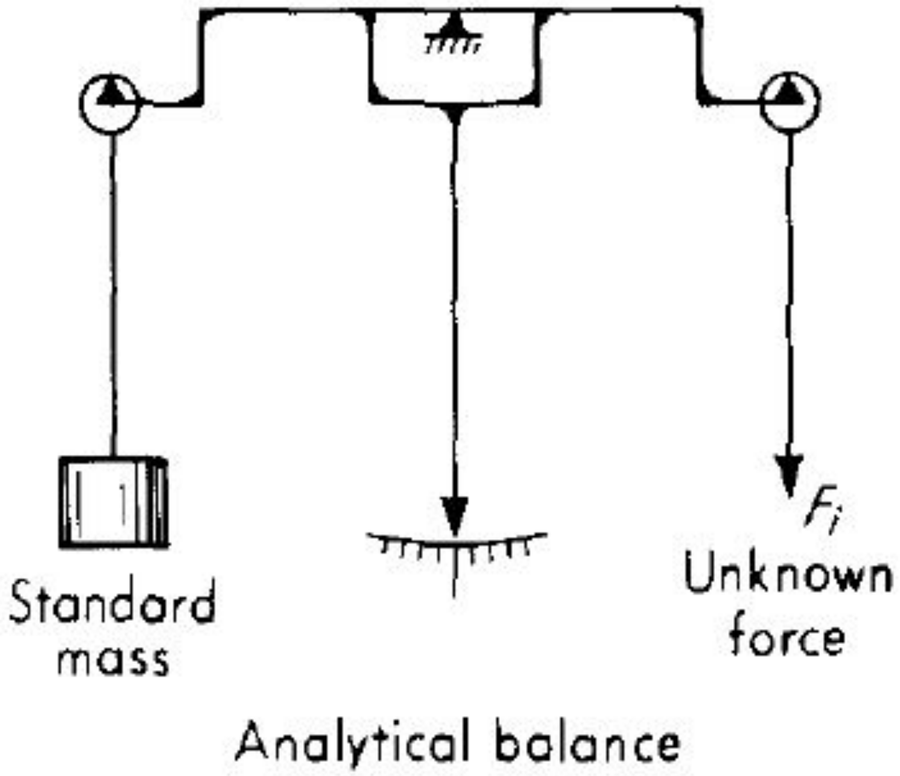
5. Applying the force to some elastic member and measuring the resulting deflection

6. Measuring the change in precession of a gyroscope caused by an applied

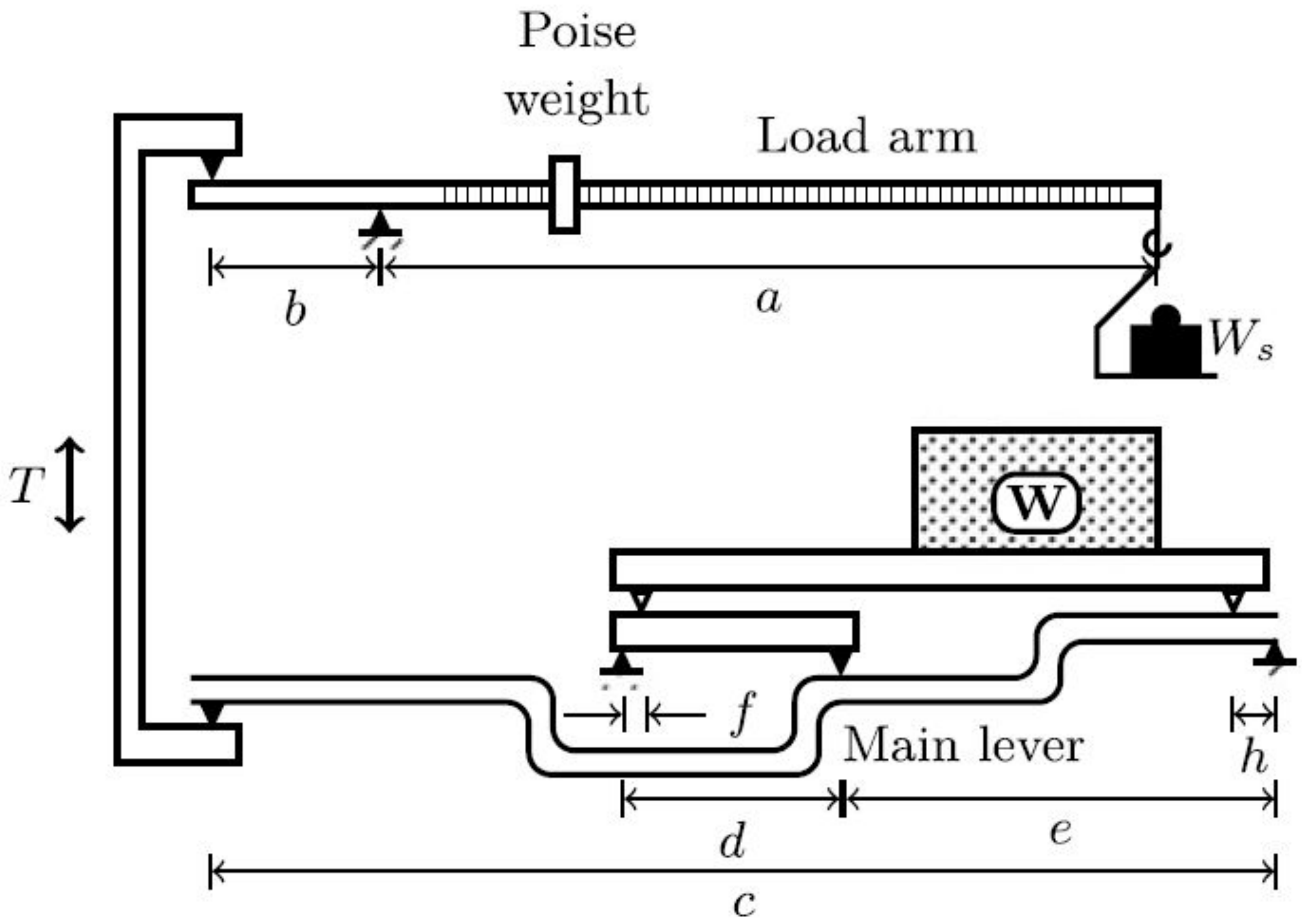
torque related to the measured force

7. Measuring the change in natural frequency of a wire tensioned by the force

# Balancing it against the known gravitational force



# Platform Balance



The weight W to be measured may be placed anywhere on the platform. The knife edges on which the platform rests share this load as W1 and W2. Let T be the force transmitted by the vertical link as shown in the figure. There are essentially four levers and the appropriate Moment equations are given below.

(1) Consider the horizontal load arm. Taking moments about the fixed fulcrum, we have

$$Tb = W_s a \qquad 1$$

(2) For the main lever, we balance the moments at the fixed fulcrum to get

$$Tc = W_2 h + W_1 \frac{f}{d} e \text{ or } T = \frac{e}{c}\left[W_2 \frac{h}{e} + W_1 \frac{f}{d}\right]$$

(3) The ratio h/e is chosen equalto f/d so that the above equation becomes

$$T = \frac{h}{c}(W_1 + W_2) = \frac{h}{c}W \qquad 3$$

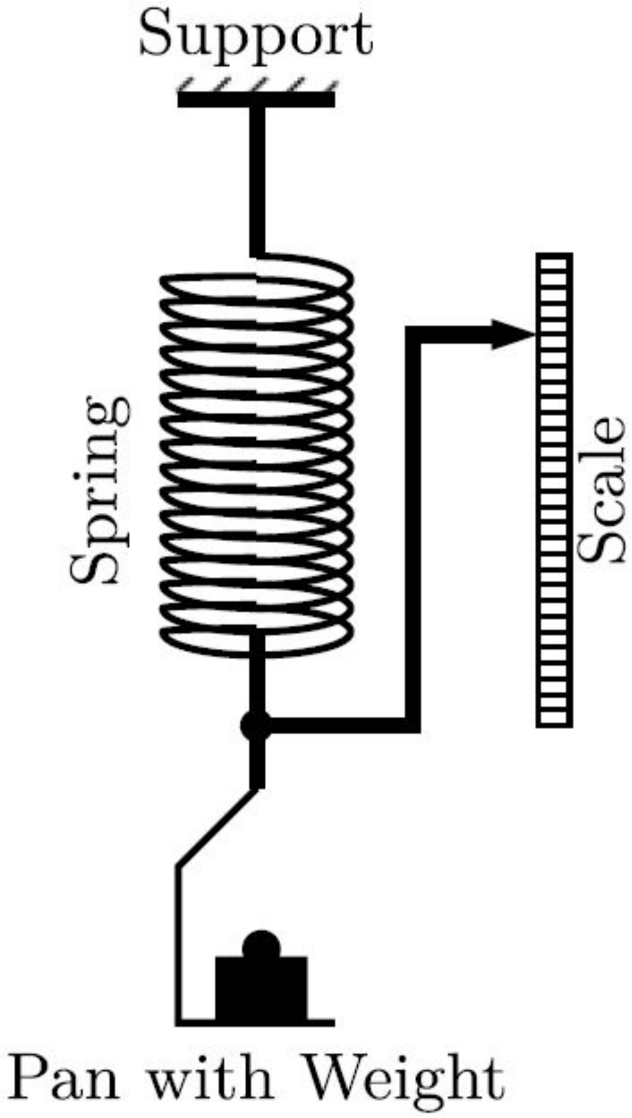
(4) Thus, T the force transmitted through the vertical link is independent of where the load is placed on the platform. From Equations in 1 and 3,

$$\boxed{T = \frac{a}{b} W_s = \frac{h}{c} W \text{ or } W = \frac{ac}{bh} W_s}$$

Thus, the gauge factor for the platform balance is ac / bh such that $W = GW_s$. In practice, the load arm floats between two stops, and the weighing is done by making the arm stay between the two stops, aligned with an etched mark. The main weights are added into the pan (usually hooked on to the end of the arm) and small Poise weight is moved along the arm to make fine adjustments. Obviously, the Poise weight $W_p$, if moved by a unit distance along the arm, is equivalent to an extra weight of $W_p$ a added in the pan. The unit of distance on the arm on which the Poise weight slides is marked in this unit!

# Force to Displacement Conversion

A spring balance (or a spring scale) is an example where a force may be converted to a displacement based on the spring constant. For a spring element (it need not actually be a spring in the form of a coil of wire), the relationship between force F and displacement x is linear and given by , F = $Kx$ ,  where K is the spring constant.


Support
Spring
Scale
Pan with Weight

The simplest device of this type is in fact the spring balance whose schematic is shown in Fig. The spring is fixed at one end and at the other end hangs a pan. The object to be weighed is placed in the pan and the position of the needle along the graduated scale gives the weight of the object. For a coiled spring like the one shown in the illustration, the spring constant is given by

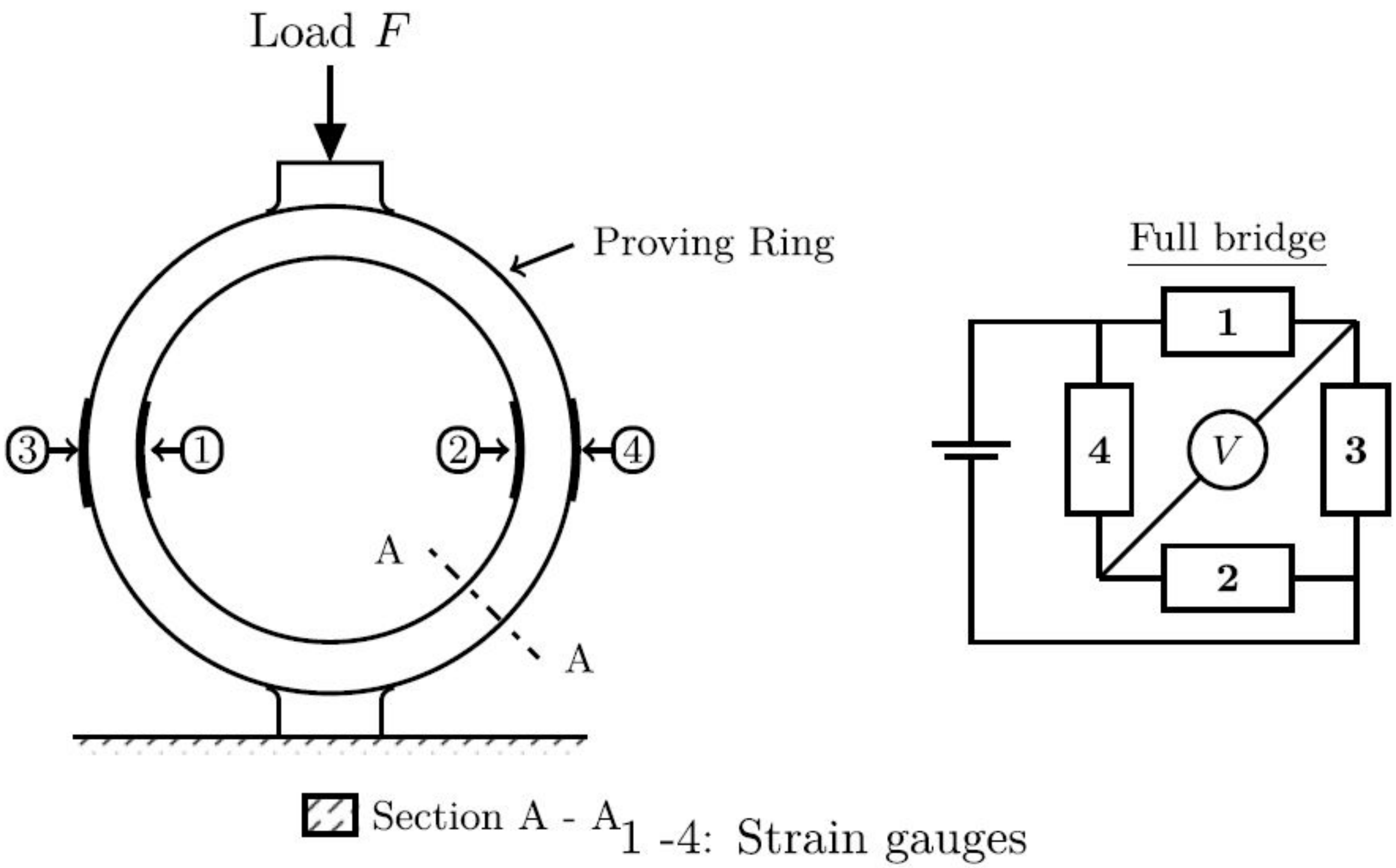
$$\boxed{K = \frac{E_s D_w^4}{8 D_m^3 N}}$$

Es - shear modulus of the material of the spring,
Dw - diameter of the wire from which the spring is wound,
Dm - mean diameter of the coil, and
N - number of coils in the spring.

# Proving Ring

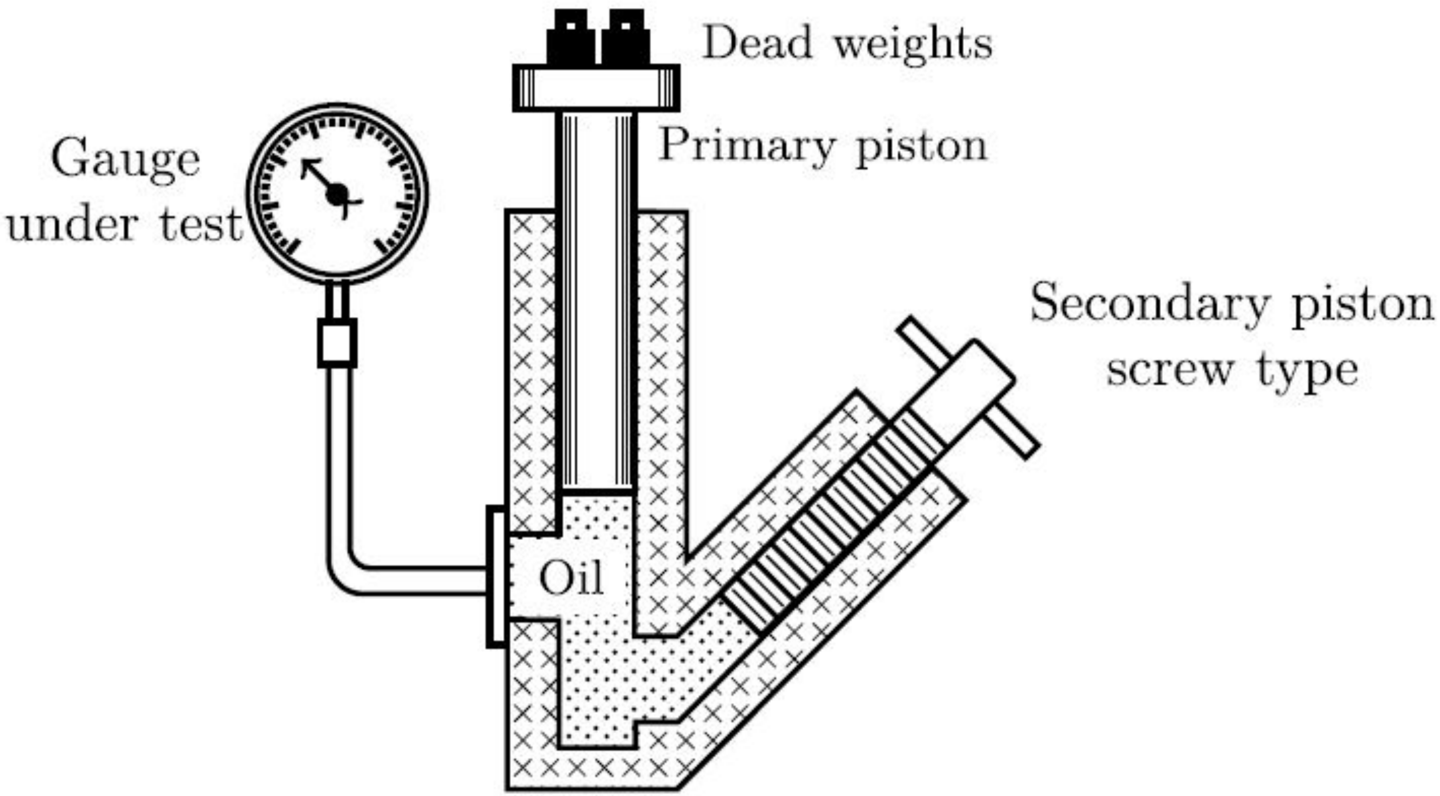
Proving ring is a device that is useful in measuring force or load. It was developed at the National Institute of Standards and Technology (NIST), USA. Proving rings are available for measuring forces in the 2–2000 kN range.

The proving ring consists of a metal ring made of alloy steel with external bosses to facilitate the application of load or force to be measured across a diameter. It has a rectangular section as shown by the sectional view across AA. Under a compressive load, the ring changes its shape with a negative strain of the inner periphery of the ring and a positive strain of the outer surface of the ring. Thus, the strain gauges 1 and 2 suffer reduction in resistance while strain gauges 3 and 4 suffer an increase in resistance. If the four gauges are connected in the form of a full bridge as shown in Fig, the bridge output is proportional to the applied load. Thus, the imbalance of the bridge is amplified by a factor of two as compared to a “half bridge” arrangement.


Load $F$
Proving Ring
3
1
2
4
A
A
Full bridge
1
4
V
3
2
Section A - A
1 -4: Strain gauges

# Conversion of Force to Hydraulic Pressure
# **Dead Weight Tester**

The weight tester consists of an arrangement by which a piston may be allowed to float over a liquid (usually oil), under internal pressure and a force in the opposite direction, imposed on the piston by weights placed as indicated in the figure. The oil pressure is changed by screwing in the piston. The pressure is calculated as the weight placed on the piston divided by the cross-sectional area of the piston (the piston is to be oriented with its axis vertical!). The gauge under test experiences the same pressure by being connected to a side tube communicating with the oil.


Dead weights
Primary piston
Gauge
under test
Secondary piston
screw type
Oil

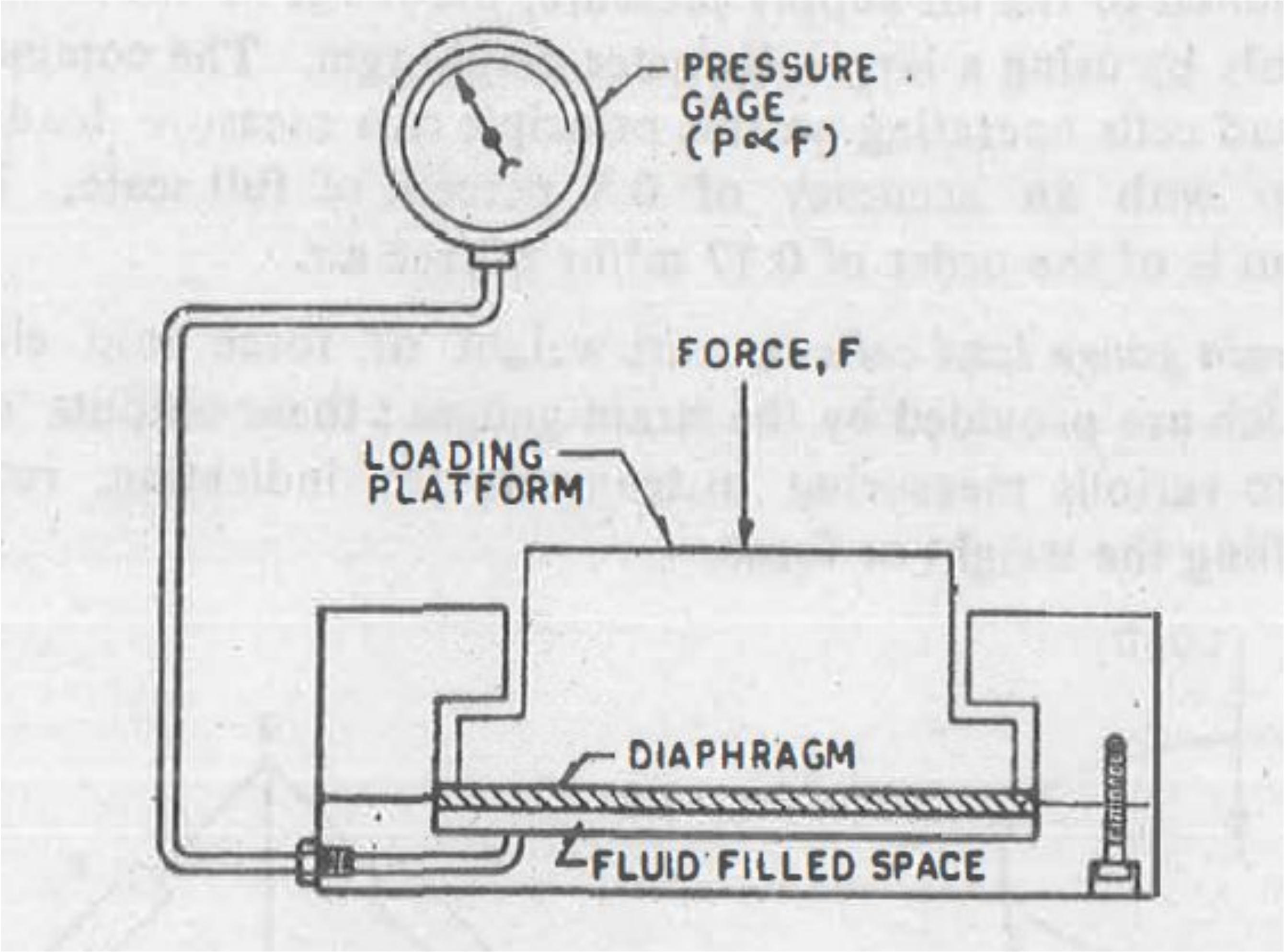
# Mechanical loall cells

The term load cell is used to decribe a variety of force transducers which may utilize the deflection or strain of elastic member, or the increase in pressure of enclosed fluids. The resulting fluid pressure is transmitted to some form of pressure sensing device such as a manometer or a bourdon tube pressure gauge. The gauge reading is identified and calibrated in units of force.

# hydraulic load cell

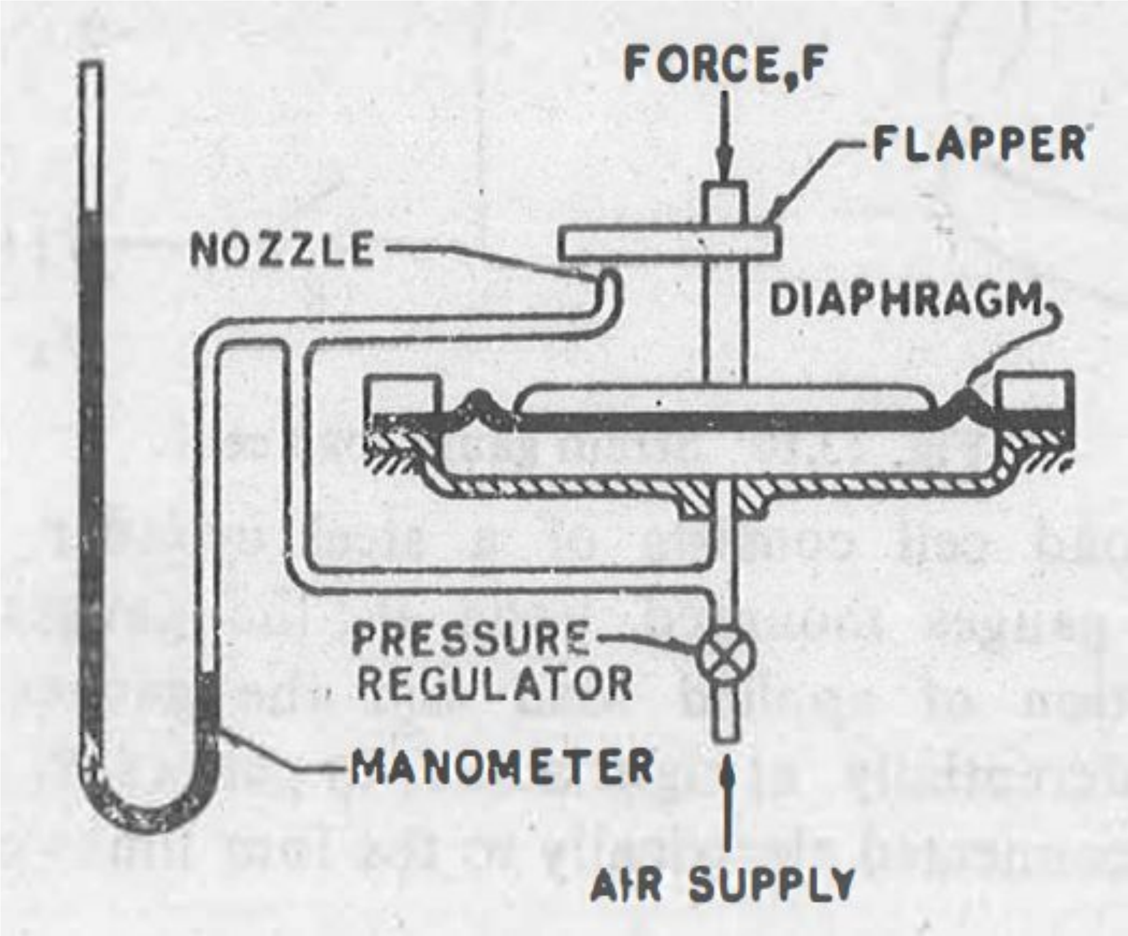
In a hydraulic load cell the force variable is impressed up on a diaphragm which deflects and thereby transmits the force to a liquid. The liquid medium, contained in a confined space, has a preload pressure of the order of 2 $kgf/cm^2$ . Application of force increases the liquid pressure ; it equals the force magnitude divided by the effective area of the diaphragm. The pressure is transmitted to and read on an accurate pressure gauge calibrated directly in force units.

# hydraulic load cell



# pneumatic load cell

A pneumatic load cell , operates on the force-balance principle and employs a nozzle-flapper transducer similar to the conventional relay system. A variable downward force is balanced by an upward force of air pressure against the effective area of a diaphragm. Application of force causes the flapper to come closer to the nozzle, and the diaphragm to deflect downwards. The nozzle opening is nearly shut-off and this results into an increased back pressure in the system. The increased pressure acts on the diaphragm, produces an effective upward force which tends to return the diaphragm to its preload position. For any constant applied force, the system attains equilibrium at a specific nozzle opening and a correspondiug pressure is indicated by the height of mercury column in a manometer. Since the maximum pressure in the system is limited to the air-supply pressure , the range of the unit can be extended only by using a larger diameter diaphragm·


FORCE,F
FLAPPER
NOZZLE
DIAPHRAGM
PRESSURE
REGULATOR
MANOMETER
AIR SUPPLY

# Strain gauge load cells

The strain gauge load cells convert weight or force into electrical

outputs which are provided by the strain gauges ; these outputs can
·be

connected to various measuring instruments for indicating, recording

and controlling the weight or force.